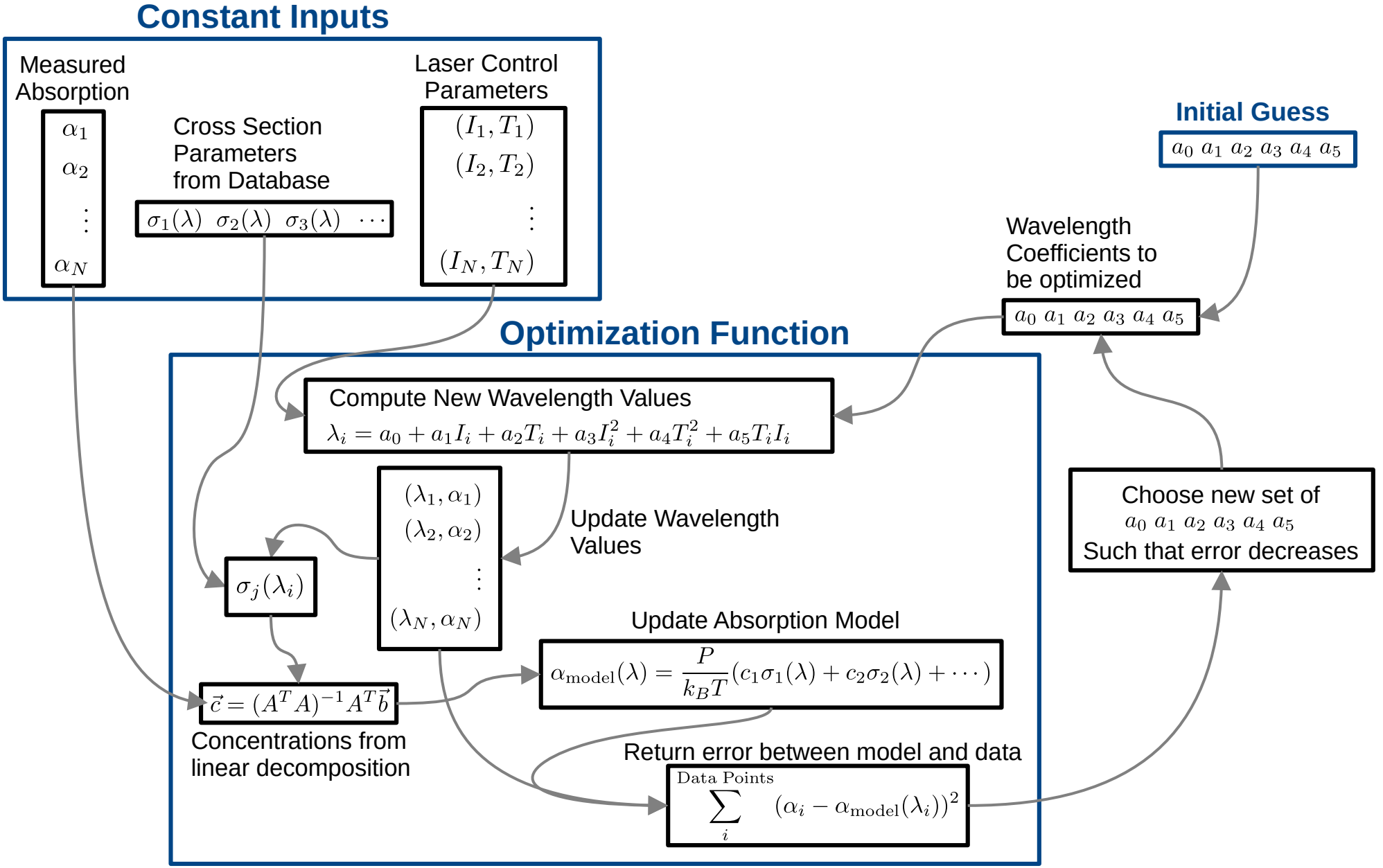

Constant Inputs
Measured Absorption
α_1
α_2
$\vdots$
α_N
Cross Section Parameters from Database
$\sigma_1(\lambda)$ $\sigma_2(\lambda)$ $\sigma_3(\lambda)$ $\cdots$
Laser Control Parameters
(I_1, T_1)
(I_2, T_2)
$\vdots$
(I_N, T_N)
Initial Guess
a_0 a_1 a_2 a_3 a_4 a_5
Wavelength Coefficients to be optimized
a_0 a_1 a_2 a_3 a_4 a_5
Optimization Function
Compute New Wavelength Values
$\lambda_i = a_0 + a_1 I_i + a_2 T_i + a_3 I_i^2 + a_4 T_i^2 + a_5 T_i I_i$
(λ_1, α_1)
(λ_2, α_2)
$\vdots$
(λ_N, α_N)
Update Wavelength Values
$\sigma_j(\lambda_i)$
$\vec{c} = (A^T A)^{-1} A^T \vec{b}$
Concentrations from linear decomposition
Update Absorption Model
$\alpha_{\text{model}}(\lambda) = \frac{P}{k_B T}(c_1\sigma_1(\lambda) + c_2\sigma_2(\lambda) + \cdots)$
Return error between model and data
Data Points
$\sum_i (\alpha_i - \alpha_{\text{model}}(\lambda_i))^2$
Choose new set of
a_0 a_1 a_2 a_3 a_4 a_5
Such that error decreases